

(12) **United States Patent**
Nishi et al.

(10) **Patent No.:** **US 12,409,562 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **METHOD FOR CONTROLLING ROBOT AND SYSTEM**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 134 days.

(21) Appl. No.: **18/408,963**

(22) Filed: **Jan. 10, 2024**

(65) **Prior Publication Data**
US 2024/0261980 A1 Aug. 8, 2024

(30) **Foreign Application Priority Data**
Feb. 6, 2023 (JP) 2023-016326

(51) **Int. Cl.**
B25J 13/02 (2006.01)
B25J 9/16 (2006.01)
B25J 13/00 (2006.01)
B25J 13/08 (2006.01)

(52) **U.S. Cl.**
CPC **B25J 13/02** (2013.01); **B25J 9/161** (2013.01); **B25J 13/006** (2013.01); **B25J 9/163** (2013.01); **B25J 13/085** (2013.01)

(58) **Field of Classification Search**
CPC . B25J 13/02; B25J 9/161; B25J 13/006; B25J 9/163; B25J 13/085; B25J 9/1602
USPC 700/245–264; 318/568.11–568.25
See application file for complete search history.

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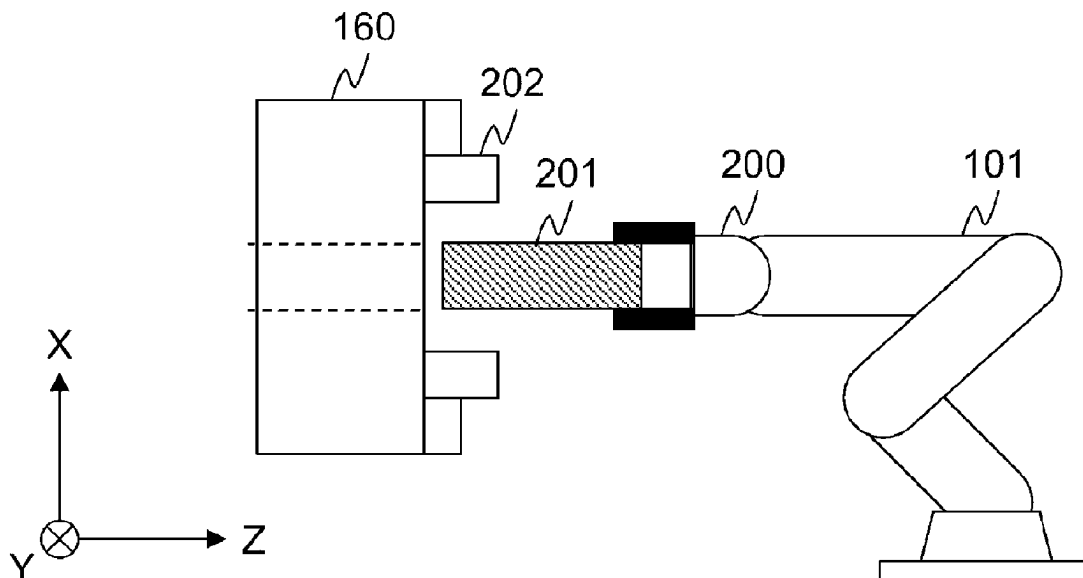
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(57) **ABSTRACT**

The computer is connected to a robot including a hand which grips a workpiece, and a machining device including a gripping mechanism which grips the workpiece, acquires first operation information regarding an operation state of the robot and second operation information regarding an operation state of the machining device; and controls the robot to cause the robot to grip the workpiece and install the workpiece in the gripping mechanism. In the control of the robot, the computer determines whether or not to start correction processing of correcting a position and an orientation of the workpiece with respect to the gripping mechanism based on the second operation information, executes the correction processing using the first operation information and the second operation information, and determines whether or not to end the correction processing based on the second operation information.

15 Claims, 14 Drawing Sheets



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FIG. 1

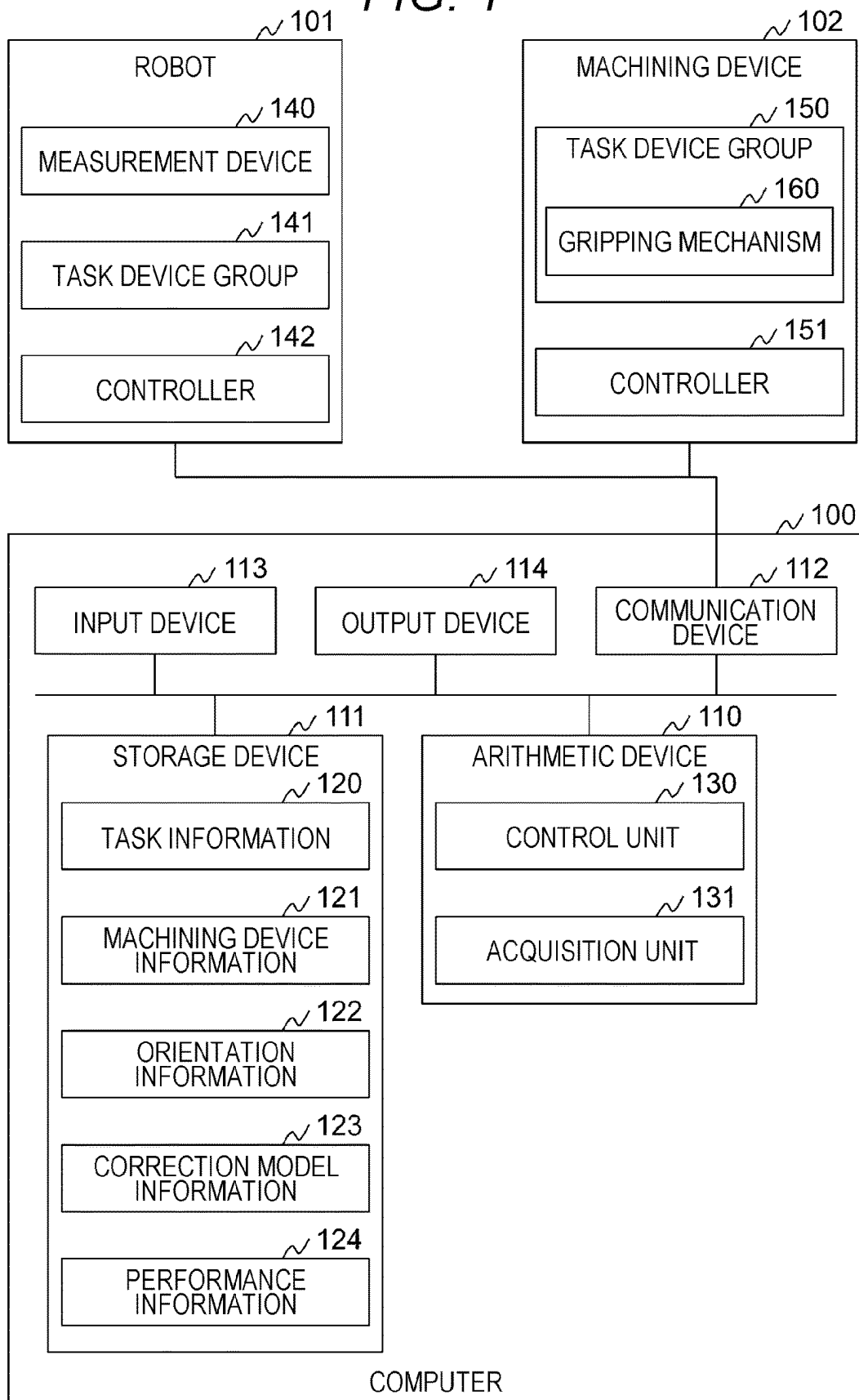


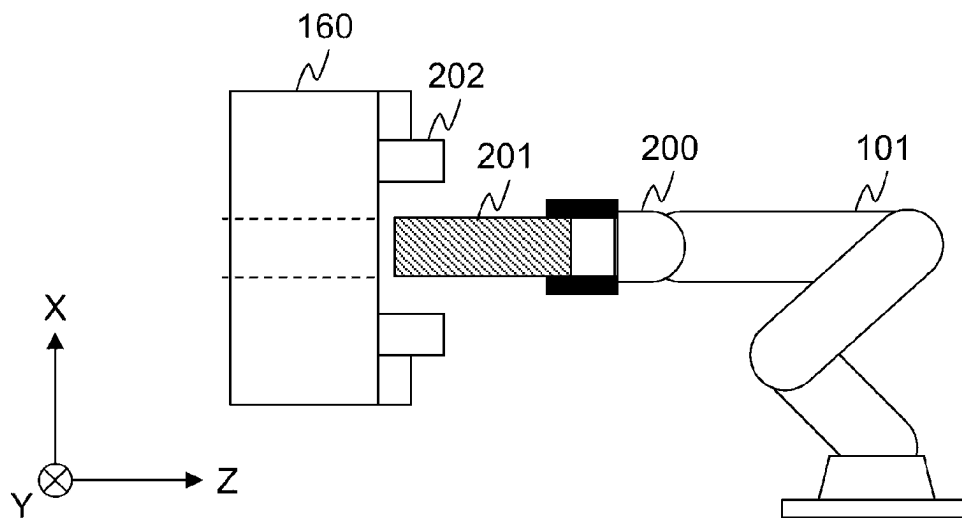
FIG. 2

FIG. 3

301 TASK NAME	302 CORRECTION MODEL NAME	303 TASK DESCRIPTION
MACHINING DEVICE A WORKPIECE A SET	WORKPIECE SET CORRECTION MODEL	OPEN HAND → MOVE WORKPIECE A TO GRIPPING ORIENTATION → CLOSE HAND → MOVE MACHINING DEVICE A TO WORKPIECE SET ORIENTATION → EXECUTE CORRECTION PROCESSING → OPEN HAND → MOVE TO INITIAL ORIENTATION
MACHINING DEVICE B WORKPIECE A RELEASE	WORKPIECE RELEASE CORRECTION MODEL	OPEN HAND → MOVE MACHINING DEVICE B TO WORKPIECE RELEASE ORIENTATION → CLOSE HAND → EXECUTE CORRECTION PROCESSING → MOVE WORKPIECE A TO RELEASE ORIENTATION → OPEN HAND → MOVE TO INITIAL ORIENTATION
⋮	⋮	⋮
MACHINING DEVICE N WORKPIECE N SET	CORRECTION MODEL	OPEN HAND → MOVE WORKPIECE N TO GRIPPING ORIENTATION → CLOSE HAND → MOVE MACHINING DEVICE N TO WORKPIECE SET ORIENTATION → EXECUTE CORRECTION PROCESSING → OPEN HAND → MOVE TO INITIAL ORIENTATION

FIG. 4

401 ~	402 ~	403 ~	404 ~	405 ~	121 ~
MACHINING DEVICE NAME	OPENING/CLOSING TASK TYPE	ELAPSED TIME	CHANGE TIME	ESTIMATED OPENING/CLOSING STATE	
MACHINING DEVICE A	OPEN	0 SECONDS	1 SECONDS	OPENING	
MACHINING DEVICE B	CLOSE	30 SECONDS	2 SECONDS	CLOSED	
⋮	⋮	⋮	⋮	⋮	
MACHINING DEVICE N	CLOSE	2 SECONDS	5 SECONDS	CLOSING	

FIG. 5

501

502

503

504

122

ID	ORIENTATION NAME	HAND POSITION	HAND ORIENTATION
1	WORKPIECE A GRIPPING ORIENTATION	(50, 0, 0)	(0, 0, 0)
2	MACHINING DEVICE A WORKPIECE SET ORIENTATION	(150, -50, 40)	(0, 0, 90)
3	INITIAL ORIENTATION	(20, 0, 30)	(0, 0, 0)
⋮	⋮	⋮	⋮
M	MACHINING DEVICE N WORKPIECE SET ORIENTATION	(100, -20, 30)	(0, 0, 90)

FIG. 6

ID	CORRECTION MODEL NAME	POSITION/ORIENTATION CORRECTION END CONDITION	GRIPPING POSITION CORRECTION END CONDITION	INTERRUPTION CONDITION	PROCESSING TIME UPPER LIMIT
1	WORKPIECE SET CORRECTION MODEL	ESTIMATED OPENING/CLOSING STATE=CLOSED	(0,0,10)	(200,200,200)	10 SECONDS
⋮	⋮	⋮	⋮	⋮	⋮
L	WORKPIECE RELEASE CORRECTION MODEL	ESTIMATED OPENING/CLOSING STATE=CLOSED	(0,0,10)	(150,150,150)	5 SECONDS

FIG. 7

701 TASK NAME	702 POSITION/ORIENTATION CORRECTION AMOUNT	703 GRIPPING POSITION CORRECTION AMOUNT	704 PEAK VALUE	705 FINAL VALUE
MACHINING DEVICE A WORKPIECE A SET	(0.2, -1.2, 0.0, 3.1, 0.1, 1.2)	(0, 0, 3.1)	(24, 98, 5)	(2, 8, 2)
MACHINING DEVICE B WORKPIECE A RELEASE	(1.2, -0.1, 0.0, 5.2, 2.0, 0.2)	(0, 0, 1.1)	(57, 18, 3)	(7, 2, 2)
⋮	⋮	⋮	⋮	⋮
MACHINING DEVICE N WORKPIECE N RELEASE	(3.2, 0.2, 0.0, 2.2, 1.0, 2.6)	(0, 0, 2.3)	(157, 22, 4)	(4, 1, 3)

FIG. 8A

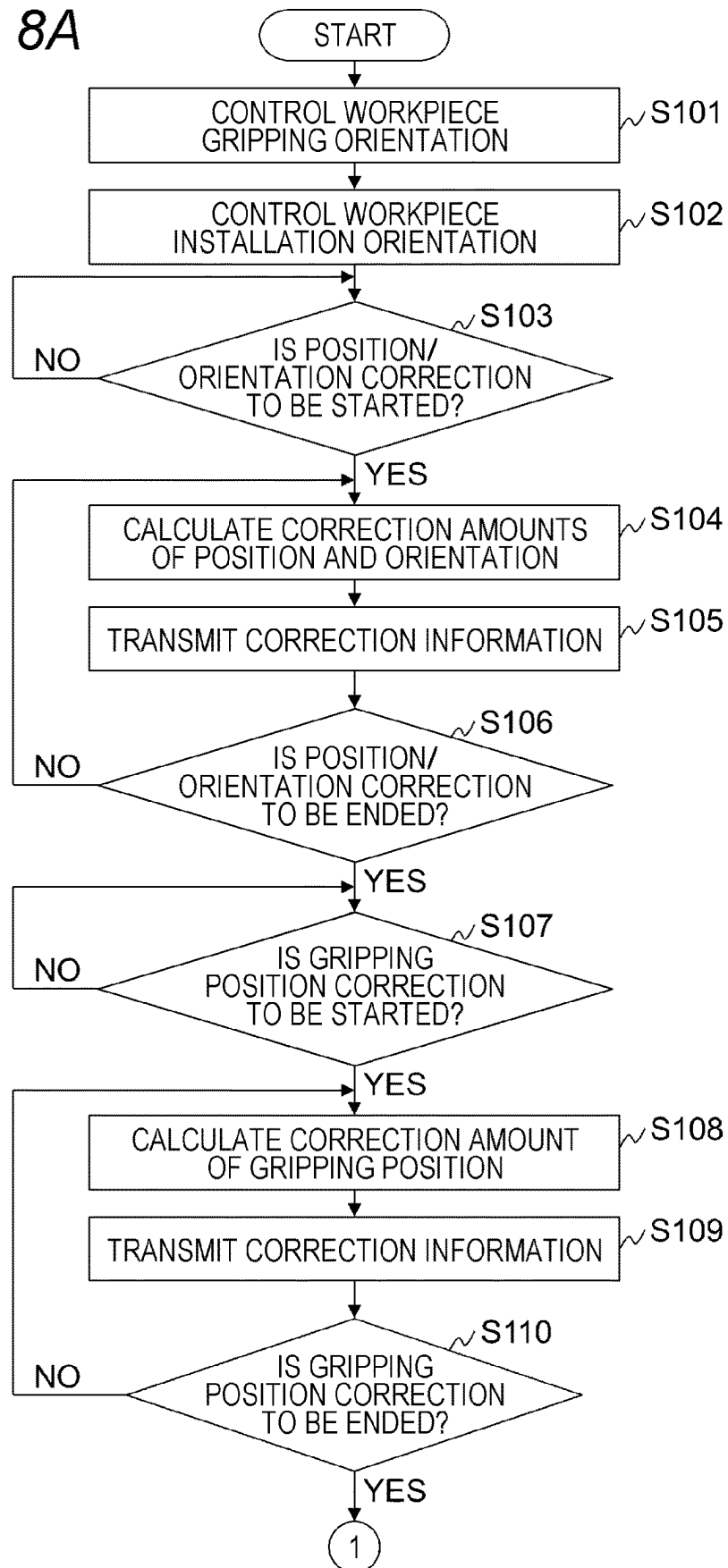


FIG. 8B

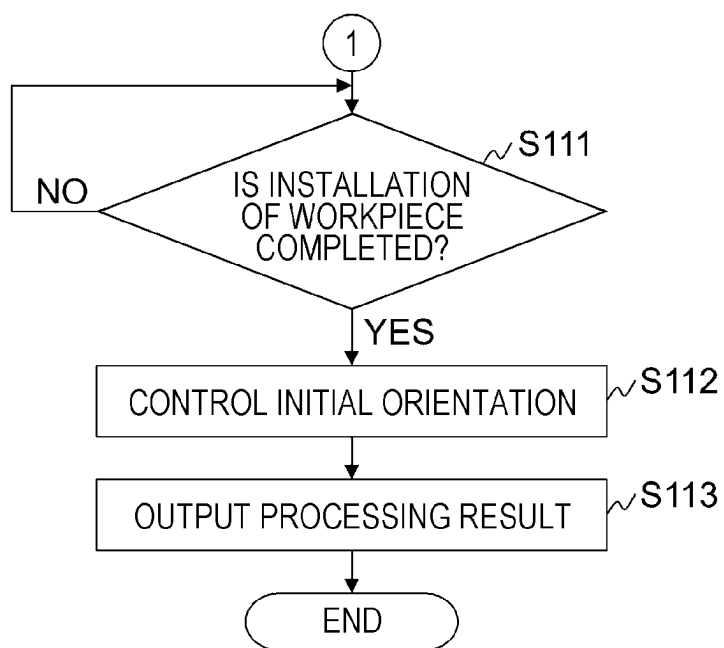


FIG. 9A

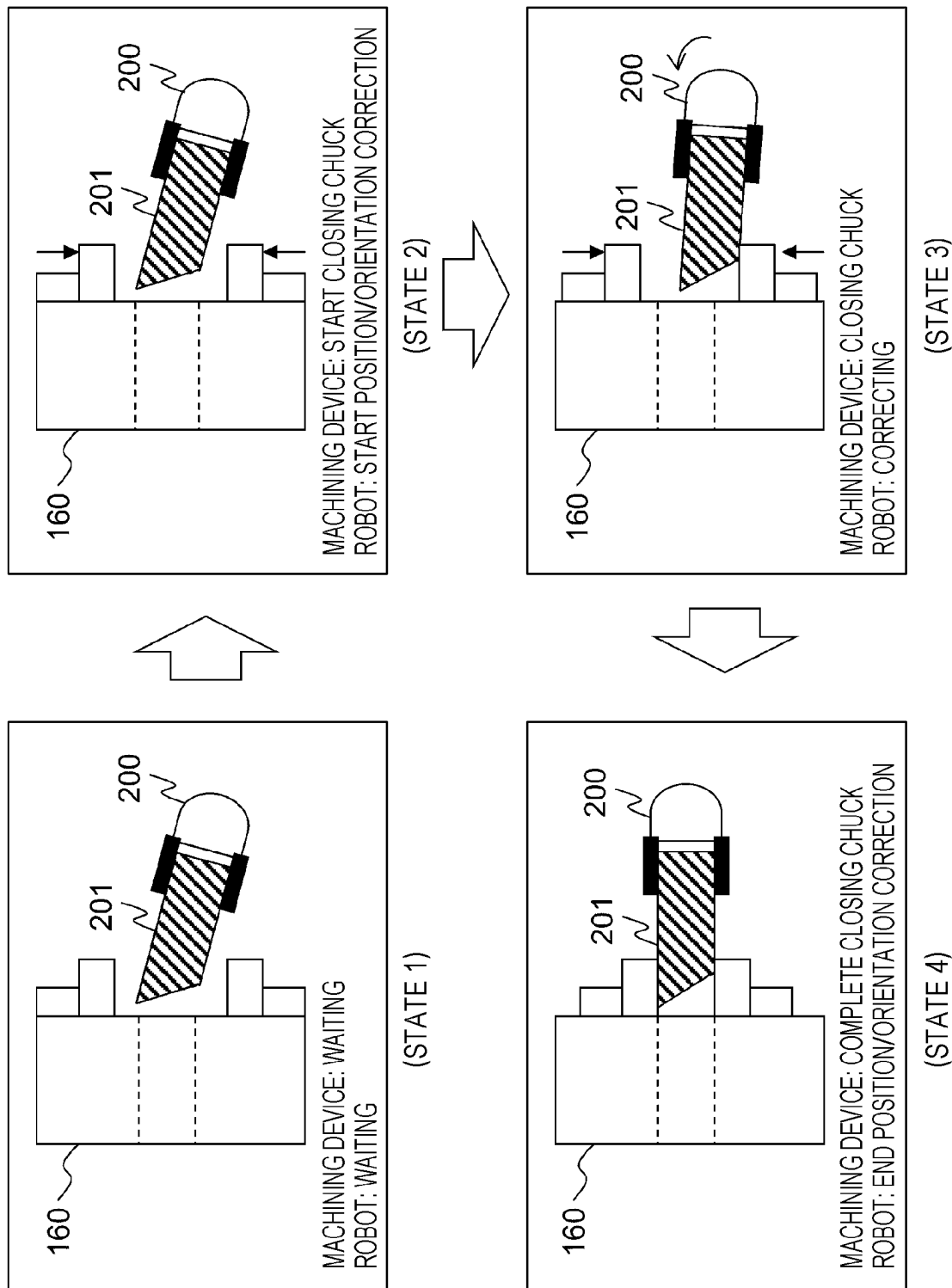


FIG. 9B

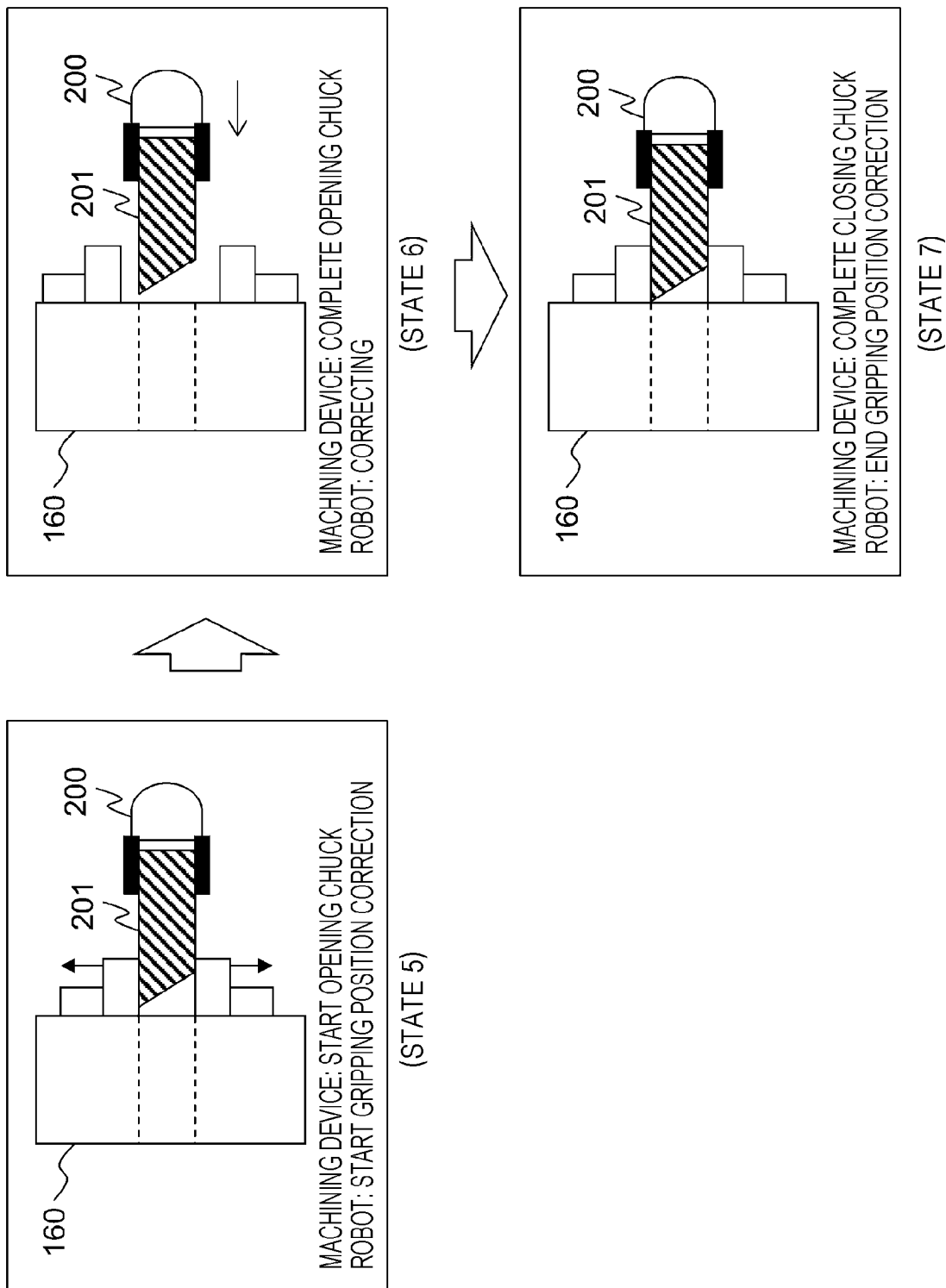


FIG. 10

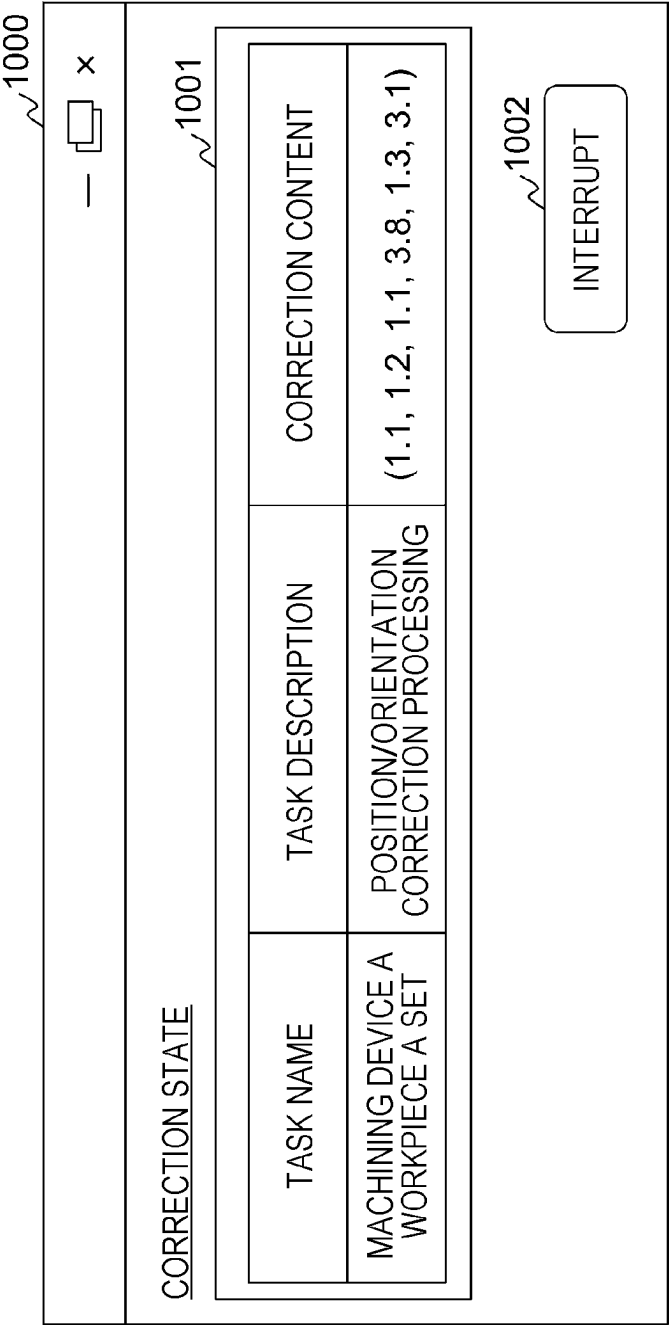
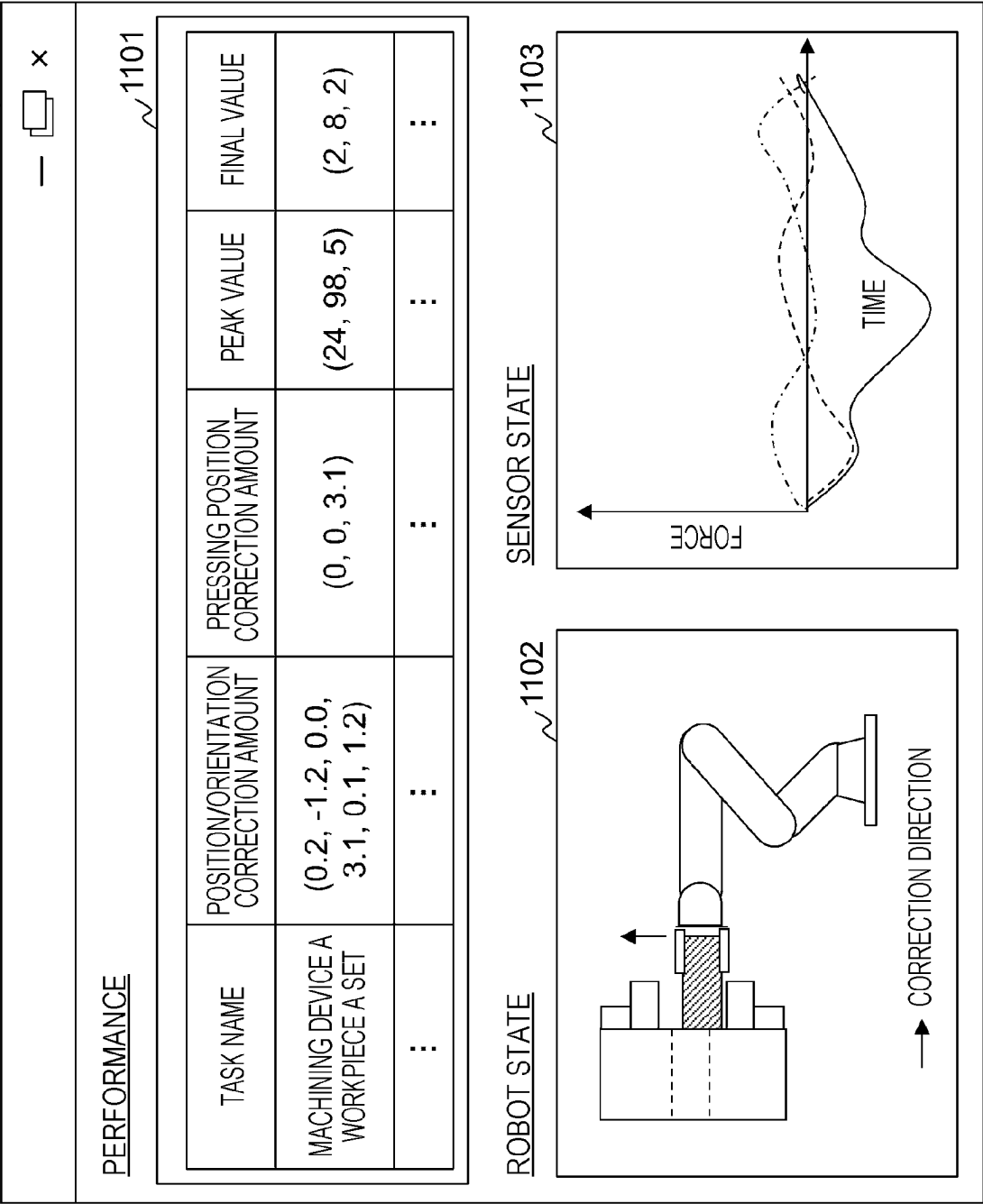


FIG. 11



ROBOT STATE

1102

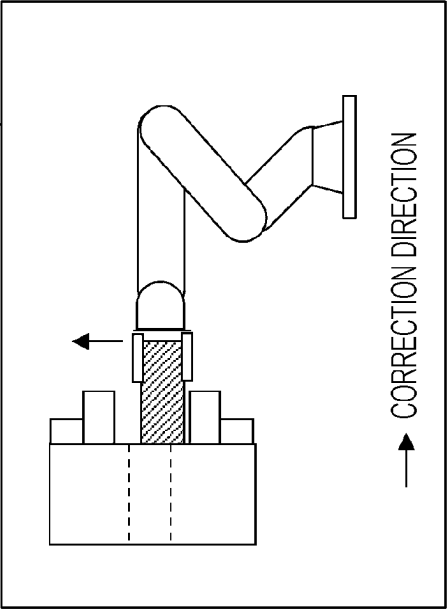


Diagram illustrating the ROBOT STATE (1102). It shows a robotic arm assembly positioned over a workpiece. A dashed line indicates the initial position, and a solid line indicates the corrected position. An arrow labeled "CORRECTION DIRECTION" points from the dashed line to the solid line, indicating the direction of adjustment.

SENSOR STATE

1103

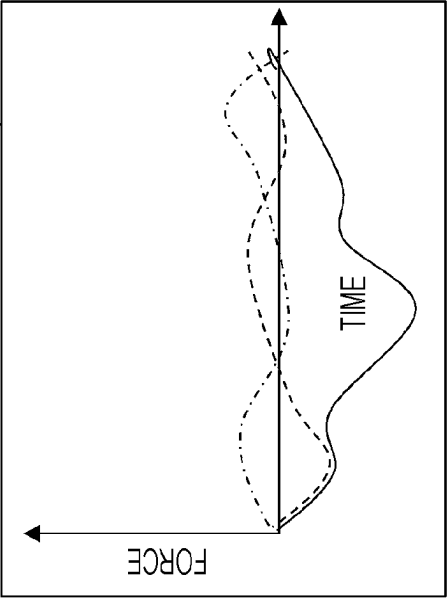


Diagram illustrating the SENSOR STATE (1103). It shows a graph of FORCE versus TIME. The solid line represents the measured force over time, showing a peak. The dashed line represents the corrected force profile, which is flatter and lower in magnitude than the measured force.

FIG. 12

601 ID	602 CORRECTION MODEL NAME	603 POSITION/ORIENTATION CORRECTION END CONDITION	604 GRIPPING POSITION CORRECTION END CONDITION	605 INTERRUPTION CONDITION	606 PROCESSING TIME UPPER LIMIT
1	WORKPIECE SET CORRECTION MODEL	ESTIMATED OPENING/CLOSING STATE=CLOSED; (10,10,10);	(0,0,10)	(200,200,200)	10 SECONDS
⋮	⋮	⋮	⋮	⋮	⋮
L	WORKPIECE RELEASE CORRECTION MODEL	ESTIMATED OPENING/CLOSING STATE=CLOSED; (10,10,10)	(0,0,10)	(150,150,150)	5 SECONDS

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METHOD FOR CONTROLLING ROBOT AND SYSTEM

CLAIM OF PRIORITY

The present application claims priority from Japanese patent application JP 2023-016326 filed on Feb. 6, 2023, the content of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to a technique for controlling a robot which performs a task of installing a workpiece in a machining device.

2. Description of the Related Art

In the manufacturing industry, a robot which installs a workpiece in a chuck mechanism of a machining device is used. The robot performs a work of installing the workpiece in the chuck mechanism based on information regarding the gripping orientation of the workpiece and the installation orientation of the workpiece with respect to the chuck mechanism.

There is a case where the installation orientation of the workpiece with respect to the chuck mechanism deviates contrary to expectation due to a tolerance of the outer shape of the workpiece, an error in the gripping orientation, an error in installation of the robot, a control error of a movable mechanism of the robot, and the like. When a large force due to the deviation is generated between the robot and the chuck mechanism, there may be problems of a failure of the robot and the chuck mechanism and a decrease in machining accuracy.

SUMMARY OF THE INVENTION

Technical Problem

A technique described in WO 2022/172873 A is known as a technique for correcting the position and orientation of the workpiece to eliminate the deviation. WO 2022/172873 A describes a system of “controlling a robot device for correcting the position and the orientation of joint of a workpiece relative to a machine tool based on the output of the force sensor” (see claim 1 of WO 2022/172873 A). Further, WO 2022/172873 A describes “A workpiece gripping mechanism is moved parallel to a vertical center line of a chuck surface of a chuck mechanism of a machine tool, an orientation of the workpiece gripping mechanism is corrected so as to reduce a force applied in a direction orthogonal to the vertical center line while pressing the workpiece against the chuck surface with a predetermined force, and a position of the workpiece gripping mechanism is corrected along the direction orthogonal to the vertical center line so as to reduce the force applied in the direction orthogonal to the vertical center line with a closing operation of the chuck mechanism.” (see claim 4 of WO 2022/172873 A).

The technique described in WO 2022/172873 A has the following problems. (1) When the shape of the end surface of the workpiece to be pressed against the chuck surface is distorted, the orientation of the workpiece after correction varies in the correction method described in WO 2022/172873 A. (2) In WO 2022/172873 A, a difference between

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the opening/closing speed of the chuck mechanism and the control speed of the robot is not considered. When the control speed of the robot is higher than the opening/closing speed of the chuck mechanism, the workpiece is separated from the claw of the chuck mechanism by the correction of the position and orientation of the workpiece by the robot. As a result, since the value of the force sensor decreases, the correction of the position and orientation of the workpiece is completed before the chuck mechanism is closed. Therefore, there is a possibility that the position and orientation of the workpiece are not correctly corrected.

The present invention provides a method for controlling a robot for correcting a position and an orientation of a workpiece to be installed in a machining device with high accuracy.

Solution to Problem

A representative example of the invention disclosed in the present application is as follows. That is, a method for controlling a robot executed by a computer, the computer including an arithmetic device, a storage device connected to the arithmetic device, and a network interface connected to the arithmetic device, the computer being connected to a robot including a hand which grips a workpiece, and a machining device including a gripping mechanism which grips the workpiece, the robot control method comprising: a first step of acquiring, by the computer, first operation information regarding an operation state of the robot and second operation information regarding an operation state of the machining device; and a second step of controlling, by the computer, the robot to cause the robot to perform a task of gripping the workpiece and installing the workpiece in the gripping mechanism, in which the second step includes: a third step of determining, by the computer, whether or not to start first correction processing of correcting a position and an orientation of the workpiece with respect to the gripping mechanism based on the second operation information, a fourth step of executing, by the computer, the first correction processing using the first operation information and the second operation information, and a fifth step of determining, by the computer, whether or not to end the first correction processing based on the second operation information.

Advantageous Effects of Invention

According to one embodiment of the present invention, the robot can correct the position and orientation of the workpiece to be installed in the machining device with high accuracy. The problems, configurations, and effects other than those described above will be clarified by the following description of embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a configuration example of a system according to a first embodiment;

FIG. 2 is a diagram illustrating an example of a task in the system according to the first embodiment;

FIG. 3 is a diagram illustrating an example of a data structure of task information according to the first embodiment;

FIG. 4 is a diagram illustrating an example of a data structure of machining device information according to the first embodiment;

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FIG. 5 is a diagram illustrating an example of a data structure of orientation information according to the first embodiment;

FIG. 6 is a diagram illustrating an example of a data structure of correction model information according to the first embodiment;

FIG. 7 is a diagram illustrating an example of a data structure of performance information according to the first embodiment;

FIG. 8A is a flowchart for describing an example of task control processing executed by a computer according to the first embodiment;

FIG. 8B is a flowchart illustrating an example of task control processing executed by the computer according to the first embodiment;

FIG. 9A is a diagram illustrating a specific example of correction processing executed by the computer according to the first embodiment;

FIG. 9B is a diagram illustrating a specific example of correction processing executed by the computer according to the first embodiment;

FIG. 10 is a diagram illustrating an example of a screen presented by the computer according to the first embodiment;

FIG. 11 is a diagram illustrating an example of a screen presented by the computer according to the first embodiment; and

FIG. 12 is a diagram illustrating an example of a data structure of correction model information according to a second embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

Hereinafter, embodiments of the present invention will be described with reference to the drawings. However, the present invention is not to be construed as being limited to the description of the following examples. Those skilled in the art can easily understand that the specific configuration can be changed without departing from the spirit or gist of the present invention.

In the configurations of the invention described below, the same or similar configurations or functions are denoted by the same reference numerals, and redundant description is omitted.

Notations such as “first”, “second”, and “third” in the present specification and the like are attached to identify components, and do not necessarily limit the number or order.

The position, size, shape, range, and the like of each configuration illustrated in the drawings and the like may not represent the actual position, size, shape, range, and the like in order to facilitate understanding of the invention. Therefore, the present invention is not limited to the position, size, shape, range, and the like disclosed in the drawings and the like.

First Embodiment

FIG. 1 is a diagram illustrating a configuration example of a system of a first embodiment. The system includes a computer 100, a robot 101, and a machining device 102. The system may include two or more computers 100, robots 101, and machining devices 102. The computer 100, the robot 101, and the machining device 102 are connected directly or via a network.

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The robot 101 performs a task such as installation of a workpiece 201 (see FIG. 2) in the machining device 102, removal of the workpiece 201 from the machining device 102, and the like, based on control information transmitted from the computer 100. The robot 101 includes a measurement device 140, a task device group 141, and a controller 142.

The measurement device 140 measures a value for acquiring the state of the robot 101 and generates measurement data. The measurement device 140 is, for example, a general sensing device such as an acceleration sensor, a force sensor, a camera, a contact sensor, a current sensor, and the like.

The task device group 141 is a device group which realizes gripping and moving of the workpiece 201, and is, for example, a general mechanical element group such as a hand, a link, a drive motor, and the like.

The controller 142 controls the task device group 141 based on the control information received from the computer 100. For example, the controller 142 moves the hand by controlling a drive motor which connects links functioning as joints according to the control information. The controller 142 transmits operation information including the angle, the angular velocity, and the angular acceleration of the joint, the torque and the current value of the drive motor, and the like to the computer 100. The controller 142 may transmit the measurement data to the computer 100 as the operation information.

The machining device 102 performs a machining task using the workpiece 201 based on the control information transmitted from the computer 100. The machining device 102 includes a task device group 150 and a controller 151.

The task device group 150 is a device group which realizes a machining task using the workpiece 201. The task device group 150 includes a gripping mechanism 160 such as a chuck. The gripping mechanism 160 grips the workpiece 201 with at least one of a point, a line, and a surface.

The controller 151 controls the task device group 150 based on the control information received from the computer 100. The controller 151 transmits operation information including information related to opening/closing control of the gripping mechanism 160, and the like to the computer 100.

The computer 100 includes an arithmetic device 110, a storage device 111, a communication device 112, an input device 113, and an output device 114. The hardware elements are connected via, for example, an internal bus, but the present invention is not limited to this method.

The input device 113 is a device which inputs data, commands, and the like to the computer 100. The input device 113 may include a keyboard, a mouse and the like. The output device 114 is a device which outputs a control value change result and the like. The output device 114 may include a display, a printer and the like. The communication device 112 is a device for communicating with an external device, and is, for example, a network interface card (NIC).

The storage device 111 is a device which stores a program and information executed by the arithmetic device 110. The storage device 111 is, for example, a hard disk drive (HDD), a solid state drive (SSD), or the like. The storage device 111 also stores information input via the input device 113 and a calculation result of the program. The storage device 111 is also used as a work area.

The storage device 111 stores task information 120, machining device information 121, orientation information 122, correction model information 123, and performance information 124.

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The task information **120** is information for managing the content of a task performed by the robot **101**. A data structure of the task information **120** will be described with reference to FIG. 3. The machining device information **121** is information for managing the state of the machining device **102**. A data structure of the machining device information **121** will be described with reference to FIG. 4. The orientation information **122** is information for managing the position and the orientation of the robot **101** in a task. A data structure of orientation information **122** will be described with reference to FIG. 5.

The correction model information **123** is information for managing a correction model which realizes correction of the position and the orientation of the workpiece **201** with respect to the gripping mechanism **160**. A data structure of the correction model information **123** will be described with reference to FIG. 6.

The performance information **124** is information for managing the performance of the correction control. A data structure of the performance information **124** will be described with reference to FIG. 7.

The arithmetic device **110** is a device which controls the computer **100**, and is, for example, a central processing unit (CPU), a graphics processing unit (GPU), a field programmable gate array (FPGA), or the like. The arithmetic device **110** executes a program stored in the storage device **111**. The arithmetic device **110** executes processing according to the program to operate as a functional unit (module) which realizes a specific function. In the following description, when processing is described with a functional unit as a subject, it indicates that the arithmetic device **110** executes a program for realizing the functional unit. The arithmetic device **110** of the first embodiment functions as a control unit **130** and an acquisition unit **131**.

The control unit **130** performs various controls related to a task. The acquisition unit **131** acquires information (operation information) related to the operation states of the robot **101** and the machining device **102**. The acquisition unit **131** acquires, for example, measurement data as operation information of the robot **101**, and acquires a control signal inside the machining device **102**, a control parameter of the task device group **150**, or the like as operation information of the machining device **102**. Note that the acquisition unit **131** may acquire the operation information from a device other than the robot **101** and the machining device **102**. For example, the acquisition unit **131** may acquire an image captured by a camera installed in the work space as the operation information of the robot **101** and the machining device **102**. In this case, the acquisition unit **131** executes information processing such as data conversion and data analysis as necessary, and specifies the operation states of the robot **101** and the machining device **102**.

Note that, regarding the functional units included in the computer **100**, a plurality of functional units may be integrated into one functional unit, or one functional unit may be divided into a plurality of functional units for each function.

FIG. 2 is a diagram illustrating an example of a task in the system according to the first embodiment. FIG. 2 illustrates the robot **101** and the machining device **102** installed in the three-dimensional space as viewed from an XZ plane.

The robot **101** grips the workpiece **201** using the hand **200**, and moves the workpiece **201** in order to install the workpiece **201** on the gripping mechanism **160**. The gripping mechanism **160** grips the workpiece **201** using claws **202** and performs machining work.

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FIG. 3 is a diagram illustrating an example of a data structure of the task information **120** according to the first embodiment.

The task information **120** stores an entry including a task name **301**, a correction model name **302**, and a task description **303**. One entry is prepared for one task. Note that the fields included in the entry are not limited to those described above. The entry may not include any of the fields described above, or may include other fields.

The task name **301** is a field for storing a name of a task. The correction model name **302** is a field for storing a name of a correction model used in a task. The task description **303** is a field for storing description of a task. The task is defined as a sequence of element tasks. The task description **303** stores a sequence of element tasks.

FIG. 4 is a diagram illustrating an example of a data structure of the machining device information **121** according to the first embodiment.

The machining device information **121** stores an entry including a machining device name **401**, an opening/closing task type **402**, an elapsed time **403**, a change time **404**, and an estimated opening/closing state **405**. One entry is prepared for one machining device **102**. Note that the fields included in the entry are not limited to those described above. The entry may not include any of the fields described above, or may include other fields.

The machining device name **401** is a field for storing a name of the machining device **102**.

The opening/closing task type **402** is a field for storing a task type of the gripping mechanism **160**. In a case where the operation information of the machining device **102** includes information indicating that the task of opening the gripping mechanism **160** has started, such as a start signal of the task of opening the gripping mechanism **160**, the acquisition unit **131** sets "open" as the opening/closing task type **402**. In a case where the operation information of the machining device **102** includes information indicating that the task of closing the gripping mechanism **160** has started, such as a start signal of the task of closing the gripping mechanism **160**, the acquisition unit **131** sets "close" in the opening/closing task type **402**.

The elapsed time **403** is a field for storing an elapsed time from the detection of the start of the task of opening the gripping mechanism **160** or the task of closing the gripping mechanism **160**.

The change time **404** is a field for storing a time (change time) required for the task of opening the gripping mechanism **160** or the task of closing the gripping mechanism **160**. The value of the change time **404** is set in advance.

The estimated opening/closing state **405** is a field for storing an estimated state of an opening/closing task of the gripping mechanism **160**. For example, in a case where the elapsed time is shorter than the change time, the acquisition unit **131** sets a value indicating that the task corresponding to the opening/closing task type **402** is being executed in the estimated opening/closing state **405**. In a case where the elapsed time is longer than the change time, the acquisition unit **131** sets a value indicating that the task corresponding to the opening/closing task type **402** is completed in the estimated opening/closing state **405**.

FIG. 5 is a diagram illustrating an example of a data structure of orientation information **122** according to the first embodiment.

The orientation information **122** stores an entry including an ID **501**, an orientation name **502**, a hand position **503**, and a hand orientation **504**. One entry is prepared for one orientation. Note that the fields included in the entry are not

limited to those described above. The entry may not include any of the fields described above, or may include other fields.

The ID **501** is a field for storing identification information of an entry. The orientation name **502** is a field for storing a name of an orientation. The task description **303** may include an element task including a name of an orientation.

The hand position **503** is a field for storing three-dimensional coordinates indicating the position of the hand **200**. The hand orientation **504** is a field for storing an angle (yaw angle, pitch angle, roll angle) indicating the orientation of the hand **200**.

In a case of performing an element task including the name of the orientation, the control unit **130** refers to the orientation information **122** to acquire information on the position and the orientation of the hand **200**. The control unit **130** generates a trajectory of the hand **200** based on the current position and orientation of the hand **200** and the acquired position and orientation of the hand. For example, the trajectory is generated using a known technique such as rapidly exploring random tree (RRT), point-to-point (PTP), and the like.

Note that information on the trajectory may be stored in advance in the orientation information **122**. In addition, the computer **100** may not hold the orientation information **122**. In this case, the computer **100** calculates the position and the orientation of the robot **101** from the position of the workpiece **201** and the like.

FIG. **6** is a diagram illustrating an example of a data structure of the correction model information **123** according to the first embodiment.

The correction model information **123** stores an entry including an ID **601**, a correction model name **602**, a position/orientation correction end condition **603**, a gripping position correction end condition **604**, an interruption condition **605**, and a processing time upper limit **606**. One entry is prepared for one correction model. Note that the fields included in the entry are not limited to those described above. The entry may not include any of the fields described above, or may include other fields.

The ID **601** is a field for storing identification information of an entry. The correction model name **602** is a field for storing a name of a correction model.

The position/orientation correction end condition **603** is a field which stores an end condition of the position and orientation correction for correcting the position/orientation of the workpiece **201** with respect to the gripping mechanism **160**. The position/orientation correction end condition **603** stores a condition related to an estimated state of the opening/closing task of the gripping mechanism **160**. In a case where the estimated opening/closing state **405** indicates "closed", the control unit **130** ends the movement orientation correction.

The gripping position correction end condition **604** is a field which stores an end condition of the gripping position correction for correcting the gripping position of the workpiece **201** of the gripping mechanism **160**. The gripping position correction end condition **604** stores thresholds of force values on the X, Y, and Z axes in the relative coordinate system of the hand **200**. In a case where the force value of each of the X axis, the Y axis, and the Z axis is greater than the thresholds set in the position/orientation correction end condition **603**, the control unit **130** ends the gripping position correction.

The interruption condition **605** is a field for storing an interruption condition of correction. The interruption condition **605** stores thresholds of force values on the X axis, the

Y axis, and the Z axis in the relative coordinate system of the hand **200**. The control unit **130** interrupts the correction in a case where the force value on any one of the X axis, the Y axis, and the Z axis is greater than the threshold set in the interruption condition **605**.

The processing time upper limit **606** is a field for storing an upper limit of the processing time required for correction. In a case where the processing time is longer than the value of the processing time upper limit **606**, the control unit **130** ends the correction.

FIG. **7** is a diagram illustrating an example of a data structure of performance information **124** according to the first embodiment.

The performance information **124** stores an entry including a task name **701**, a position/orientation correction amount **702**, a gripping position correction amount **703**, a peak value **704**, and a final value **705**. One entry is prepared for one task. Note that the fields included in the entry are not limited to those described above. The entry may not include any of the fields described above, or may include other fields.

The task name **701** is a field for storing a name of a task.

The position/orientation correction amount **702** is a field for storing a correction amount in the position/orientation correction. The position/orientation correction amount **702** stores movement amounts of the hand **200** on the X axis, the Y axis, and the Z axis, and change amounts of the yaw angle, the pitch angle, and the roll angle.

The gripping position correction amount **703** is a field for storing a correction amount in the gripping position correction. The gripping position correction amount **703** stores movement amounts of the hand **200** on the X axis, the Y axis, and the Z axis. Since the orientation of the workpiece **201** is not corrected in the gripping position correction, the gripping position correction amount **703** does not include the change amounts of the yaw angle, the pitch angle, and the roll angle.

The peak value **704** is a field for storing the maximum force value in the task. The final value **705** is a field for storing a force value in a case where the task is completed.

FIGS. **8A** and **8B** are flowcharts for explaining an example of task control processing executed by the computer **100** according to the first embodiment. FIGS. **9A** and **9B** are diagrams illustrating specific examples of correction processing executed by the computer **100** according to the first embodiment.

Here, control processing of task in which the robot **101** grips the workpiece **201** and installs the workpiece **201** in the machining device **102** will be described.

The control unit **130** controls the workpiece gripping orientation of the robot **101** based on the operation information, the task information **120**, and the orientation information **122** of the robot **101** (step **S101**). For example, the control unit **130** generates a trajectory for transition from the current position/orientation of the hand **200** to the workpiece gripping orientation, generates control information of the robot **101** based on the trajectory, and transmits the control information.

The control unit **130** controls the workpiece installation orientation of the robot **101** based on the operation information, the task information **120**, and the orientation information **122** of the robot **101** (step **S102**). For example, the control unit **130** generates a trajectory for transition from the current position and orientation of the hand **200** to the workpiece installation orientation, generates control information of the robot **101** based on the trajectory, and transmits the control information.

Here, it is assumed that the positional relationship among the hand **200**, the gripping mechanism **160**, and the workpiece **201** of the robot **101** is (state 1) in FIG. 9A. In a case where the gripping mechanism **160** is closed in the positional relationship illustrated in (state 1), a large force acts between the hand **200** and the gripping mechanism **160**, and there is a possibility that a failure or the like of the hand **200** occurs. Therefore, it is necessary to correct the position and orientation of the workpiece **201**.

The control unit **130** refers to the machining device information **121** and determines whether or not to start position/orientation correction (step **S103**).

Specifically, the control unit **130** determines whether the opening/closing task type **402** of the entry corresponding to the machining device **102** in which the workpiece **201** is installed is "closed". In a case where the opening/closing task type **402** is "closed", the control unit **130** starts position/orientation correction. That is, as illustrated in (state 2) of FIG. 9A, in a case where the gripping mechanism **160** starts the task of closing, the control unit **130** starts the position/orientation correction.

In a case where the opening/closing task type **402** is "open", the control unit **130** returns to step **S103** after a lapse of a certain period of time.

In a case where the opening/closing task type **402** is "closed", the control unit **130** starts position/orientation correction and calculates correction amounts of the position and orientation of the workpiece **201** based on the position, orientation, and force value of the robot **101** (step **S104**).

In the position/orientation correction, the insertion orientation of the workpiece **201** with respect to the gripping mechanism **160** is corrected in consideration of the force acting on the robot **101**. For example, the control unit **130** calculates the correction amounts of the position and orientation of the workpiece **201** such that the force value decreases and the center of gravity of the workpiece **201** is at a predetermined position of the gripping mechanism **160**. Here, the predetermined position of the gripping mechanism **160** means a position on the XY plane in FIG. 2.

The control unit **130** transmits the correction information including the position and orientation of the hand **200** corrected based on the correction amount to the robot **101** (step **S105**). As shown in (state 3) of FIG. 9A, the robot **101** moves and rotates the hand **200** based on the correction information.

The control unit **130** determines whether or not to end the position/orientation correction based on the operation information of the robot **101** and the correction model information **123** (step **S106**).

Specifically, the control unit **130** determines whether or not the estimated opening/closing state **405** of the entry corresponding to the machining device **102** in which the workpiece **201** is installed is "closed". That is, as illustrated in (state 4) of FIG. 9A, in a case where the gripping mechanism **160** completes the closing work, the control unit **130** ends the position/orientation correction.

In a case where the position/orientation correction end condition **603** is not satisfied, the control unit **130** returns to step **S104** and executes similar processing.

In a case where the position/orientation correction end condition **603** is satisfied, the control unit **130** refers to the machining device information **121** and determines whether or not to start correction of the gripping position (step **S107**).

Specifically, the control unit **130** determines whether the opening/closing task type **402** of the entry corresponding to the machining device **102** in which the workpiece **201** is installed is "open". In a case where the opening/closing task

type **402** is "open", the control unit **130** starts correction of the gripping position. That is, as illustrated in (state 5) of FIG. 9B, in a case where the gripping mechanism **160** starts an opening task, the control unit **130** starts to correct the gripping position.

After starting the correction of the gripping position, the control unit **130** calculates the correction amount of the gripping position of the workpiece **201** based on the position, orientation, and force value of the robot **101** (step **S108**).

In the correction of the gripping position, the depth of insertion of the workpiece **201** into the gripping mechanism **160** is corrected so that the workpiece **201** does not fall off from the gripping mechanism **160**. For example, the control unit **130** calculates the correction amount of the gripping position of the workpiece **201** so that the force value acting on the workpiece **201** by the contact of the gripping mechanism **160** is greater than a predetermined threshold, and the current position and orientation of the workpiece **201** can be maintained. Here, the insertion depth means a position in the Z direction in FIG. 2.

The control unit **130** transmits the correction information including the position and orientation of the hand **200** corrected based on the correction amount to the robot **101** (step **S109**). The robot **101** moves the hand **200** in a direction in which the workpiece **201** is pushed into the gripping mechanism **160** as illustrated in (state 6) of FIG. 9B based on the correction information.

The control unit **130** determines whether or not to end the correction of the gripping position based on the operation information of the robot **101** and the correction model information **123** (step **S110**).

Specifically, the control unit **130** determines whether the force values on the X axis, the Y axis, and the Z axis in the relative coordinate system of the hand **200** are greater than the value of the position/orientation correction end condition **603**. That is, as illustrated in (state 7) of FIG. 9B, in a case where the force of pressing the workpiece **201** against the gripping mechanism **160** is greater than the threshold, that is, in a case where the workpiece **201** is inserted sufficiently deep into the gripping mechanism **160**, the control unit **130** ends the gripping position correction.

In a case where the interruption condition is satisfied or the execution time is longer than the processing time upper limit **606** during the execution of the position/orientation correction, the control unit **130** interrupts the correction and outputs an error. Similar control is performed for the gripping position correction.

The control unit **130** determines whether the installation of the workpiece **201** in the gripping mechanism **160** is completed (step **S111**).

Specifically, the control unit **130** determines whether or not the estimated opening/closing state **405** of the entry corresponding to the machining device **102** in which the workpiece **201** is installed is "closed". In a case where the estimated opening/closing state **405** is "closed", the control unit **130** determines that the installation of the workpiece **201** in the gripping mechanism **160** is completed.

In a case where the installation of the workpiece **201** in the gripping mechanism **160** is not completed, the control unit **130** returns to step **S111** after a lapse of a certain time.

In a case where the installation of the workpiece **201** in the gripping mechanism **160** is completed, the control unit **130** performs control for returning the robot **101** to the initial orientation (step **S112**). Furthermore, the control unit **130** outputs the processing result (step **S113**), and ends the processing.

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FIGS. 10 and 11 are diagrams illustrating an example of a screen presented by the computer 100 according to the first embodiment.

A screen 1000 is a screen for presenting information related to correction being executed. The screen 1000 includes a display field 1001 and an interruption button 1002.

In the display field 1001, the name of the task to be corrected, the type of correction, the correction amount, and the like are displayed. The interruption button 1002 is an operation button for instructing interruption of correction.

A screen 1100 is a screen for presenting information related to actual performance of correction. The screen 1100 includes display fields 1101, 1102, and 1103.

The display field 1101 is a field for displaying the performance information 124. Only a part of the performance information 124 may be displayed in the display field 1101. The display field 1102 is a field for displaying an image for reproducing the movement of the robot 101 by the correction. The display field 1103 is a field for displaying a graph indicating a force value at the time of task.

The position/orientation correction can suppress variations in the position and orientation of the workpiece 201 even in a case where the shape of the end surface of the workpiece 201 in contact with the gripping mechanism 160 is distorted. Therefore, the workpiece 201 can be installed on the gripping mechanism 160 at the correct position and orientation.

Furthermore, the control unit 130 controls the start and end of the position/orientation correction based on the opening/closing state of the gripping mechanism 160. As a result, the correction can be performed without being affected by the contact state between the workpiece 201 and the gripping mechanism 160 and the difference in operating speed between the robot 101 and the gripping mechanism 160. Therefore, the accuracy of correction can be improved.

In the gripping position correction, the workpiece 201 can be prevented from falling off from the gripping mechanism 160 by moving the workpiece 201 so as to be pushed into the gripping mechanism 160.

Furthermore, the control unit 130 controls the start of the gripping position correction based on the opening/closing state of the gripping mechanism 160, and controls the end of the gripping position correction based on the force acting on the robot 101. As a result, the correction can be performed without being affected by the contact state between the workpiece 201 and the gripping mechanism 160 and the difference in operating speed between the robot 101 and the gripping mechanism 160. Therefore, the accuracy of correction can be improved.

As described above, the computer 100 according to the first embodiment can correct the position and orientation of the workpiece 201 to be installed in the machining device 102 with high accuracy by performing the position/orientation correction and the gripping position correction.

Second Embodiment

In the second embodiment, a correction end condition is different from that in the first embodiment. Hereinafter, the second embodiment will be described focusing on a difference from the first embodiment.

The configuration of the system of the second embodiment is the same as that of the first embodiment. The task information 120, the machining device information 121, the

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orientation information 122, and the performance information 124 of the second embodiment are the same as those of the first embodiment.

In the second embodiment, the content of the correction model information 123 is partially different. FIG. 12 is a diagram illustrating an example of a data structure of the correction model information 123 according to the second embodiment.

The structure of the entry stored in the correction model information 123 is the same as that of the first embodiment. The second embodiment is different from the first embodiment in the content set in the position/orientation correction end condition 603.

The position/orientation correction end condition 603 stores the opening/closing state of the gripping mechanism 160 and thresholds of the X axis, Y axis, and Z axis forces in the relative coordinate system of the hand 200. The control unit 130 ends the movement orientation correction in a case where the force value of each of the X axis, the Y axis, and the Z axis is smaller than the threshold set in the position/orientation correction end condition 603 and the estimated opening/closing state 405 is "closed".

The task control processing executed by the computer 100 of the second embodiment is partially different from that of the first embodiment. Specifically, a method of determining the end of the position/orientation correction is different.

In addition, in a case where the gripping mechanism 160 is already closed in the determination of the end of the position/orientation correction, the control unit 130 instructs the machining device 102 to open the gripping mechanism 160 and close the gripping mechanism after a certain period of time has elapsed.

According to the second embodiment, position/orientation correction with higher accuracy can be realized.

Note that the present invention is not limited to the above-described embodiments, and includes various modifications. For example, the above-described embodiments have been described in detail in order to simply describe the present invention, and are not necessarily limited to those having all the described configurations. In addition, it is also possible to add, delete, and replace other configurations for a part of the configuration of each embodiment.

In addition, a part or all of the above-described configurations, functions, processors, processing means, and the like may be realized by hardware, for example, by designing with an integrated circuit. In addition, the present invention can also be realized by a program code of software that realizes the functions of the embodiments. In this case, a storage medium in which the program code is recorded is provided to a computer, and a processor included in the computer reads the program code stored in the storage medium. In this case, the program code itself read from the storage medium realizes the functions of the above-described embodiments, and the program code itself and the storage medium storing the program code constitute the present invention. As a storage medium for supplying such a program code, for example, a flexible disk, a CD-ROM, a DVD-ROM, a hard disk, a solid state drive (SSD), an optical disk, a magneto-optical disk, a CD-R, a magnetic tape, a non-volatile memory card, a ROM, or the like is used.

In addition, the program code for realizing the functions described in the present embodiment can be implemented by a wide range of programs or script languages such as assembler, C/C++, perl, Shell, PHP, Python, Java (registered trademark), and the like.

Furthermore, a program code of software that realizes the functions of the embodiments may be distributed via a

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network to be stored in a storage means such as a hard disk or a memory of a computer or a storage medium such as a CD-RW or a CD-R, and a processor included in the computer may read and execute the program code stored in the storage means or the storage medium.

In addition, the control lines and the information lines indicate those necessary for the description, and do not necessarily indicate all the control lines and the information lines on the product. All the configurations may be connected to each other.

What is claimed is:

1. A method for controlling a robot executed by a computer,

the computer including

an arithmetic device, a storage device connected to the arithmetic device, and a network interface connected to the arithmetic device,

the computer being connected to a robot including a hand which grips a workpiece, and a machining device including a gripping mechanism which grips the workpiece,

the method for controlling the robot comprising:

a first step of acquiring, by the computer, first operation information regarding an operation state of the robot and second operation information regarding an operation state of the machining device; and

a second step of controlling, by the computer, the robot to cause the robot to perform a task of gripping the workpiece and installing the workpiece in the gripping mechanism,

wherein the second step includes:

a third step of determining, by the computer, whether or not to start first correction processing of correcting a position and an orientation of the workpiece with respect to the gripping mechanism based on the second operation information;

a fourth step of executing, by the computer, the first correction processing using the first operation information and the second operation information; and

a fifth step of determining, by the computer, whether or not to end the first correction processing based on the second operation information.

2. The method of controlling the robot according to claim 1, wherein

the third step includes a step of starting, by the computer, the first correction processing in a case where a task of closing the gripping mechanism starts, and

the fifth step includes a step of ending the first correction processing in a case where the task of closing the gripping mechanism is completed.

3. The method of controlling the robot according to claim 2, wherein

the fifth step includes a step of determining, by the computer, whether or not the task of closing the gripping mechanism is completed based on an elapsed time from start of the task of closing the gripping mechanism.

4. The method of controlling the robot according to claim 3, wherein

the fifth step includes a step of ending, by the computer, the first correction processing in a case where the task of closing the gripping mechanism is completed and a force acting on the hand is smaller than a threshold value.

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5. The method of controlling the robot according to claim

1, wherein

the second step includes:

a sixth step of determining, by the computer, whether or not to start second correction processing of correcting

a gripping position of the workpiece of the gripping mechanism based on the second operation information;

a seventh step of executing, by the computer, the second correction processing using the first operation information and the second operation information; and

an eighth step of determining, by the computer, whether or not to end the second correction processing based on the first operation information.

6. The method of controlling the robot according to claim 5, wherein

the sixth step includes a step of starting, by the computer, the second correction processing in a case where a task of opening the gripping mechanism starts, and

the eighth step includes a step of ending, by the computer, the second correction processing in a case where a force of pressing the workpiece against the gripping mechanism is greater than a threshold value.

7. The method of controlling the robot according to claim

5, wherein

the computer executes the sixth step, the seventh step, and the eighth step after executing the fifth step.

8. The method of controlling the robot according to claim 7, wherein

the fourth step includes a step of interrupting, by the computer, the first correction processing in a case where a force equal to or greater than a predetermined value acts on the hand or in a case where an execution time of the first correction processing is equal to or greater than a predetermined value, and

the seventh step includes a step of interrupting, by the computer, the second correction processing in a case where a force equal to or greater than a predetermined value acts on the hand or in a case where an execution time of the second correction processing is equal to or greater than a predetermined value.

9. The method of controlling the robot according to claim 7, wherein

the gripping mechanism grips the workpiece with at least one of a point, a line, and a surface.

10. A system comprising a computer, a robot including a hand which grips a workpiece, and a machining device including a gripping mechanism which grips the workpiece, wherein the computer

acquires first operation information regarding an operation state of the robot and second operation information regarding an operation state of the machining device, and

executes a robot control processing to cause the robot to perform a task of gripping the workpiece and installing the workpiece in the gripping mechanism,

in the robot control processing, the computer executes position/orientation correction including:

first processing of determining whether or not to start first correction processing of correcting a position and an orientation of the workpiece with respect to the gripping mechanism based on the second operation information;

second processing of executing the first correction processing using the first operation information and the second operation information; and

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third processing of determining whether or not to end the first correction processing based on the second operation information.

11. The system according to claim 10, wherein in the first processing, the computer starts the first correction processing in a case where a task of closing the gripping mechanism starts, and

in the third processing, the computer ends the first correction processing in a case where the task of closing the gripping mechanism is completed.

12. The system according to claim 11, wherein in the third processing, the computer ends the first correction processing in a case where the task of closing the gripping mechanism is completed and a force acting on the hand is smaller than a threshold value.

13. The system according to claim 10, wherein in the robot control processing, the computer executes gripping position correction including:
fourth processing of determining whether or not to start second correction processing of correcting a gripping

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position of the workpiece of the gripping mechanism based on the second operation information;

fifth processing of executing the second correction processing using the first operation information and the second operation information; and

sixth processing of determining whether or not to end the second correction processing based on the first operation information.

14. The system according to claim 13, wherein in the fourth processing, the computer starts the second correction processing in a case where the task of opening the gripping mechanism starts, and

in the sixth processing, the computer ends the second correction processing in a case where a force of pressing the workpiece against the gripping mechanism is greater than a threshold value.

15. The system according to claim 13, wherein the gripping position correction is executed after execution of the position/orientation correction.

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